

Problem 97. The positive integers $a_0, a_1, a_2, \dots, a_{3030}$ satisfy

$$2a_{n+2} = a_{n+1} + 4a_n \quad \text{for } n = 0, 1, 2, \dots, 3028.$$

Prove that at least one of the numbers $a_0, a_1, a_2, \dots, a_{3030}$ is divisible by 2^{2020} .

There are different ways of solving the problem. All of these use some induction argument. Most of these proofs use one of the following two lemmas. In many places, they can be used interchangeably.

Lemma 1. If a, b, c, d are integers with $2c = b + 4a$ and $2d = c + 4b$, then $4 \mid b$.

Proof. From $2d = c + 4b$ we have that c is even, and so from $b = 2c - 4a$ it follows that $4 \mid b$.

Lemma 2. For $0 \leq n \leq 3030$, denote by v_n the largest integer such that 2^{v_n} divides a_n . We claim the following:

$$v_{n+1} \geq \min(v_n + 2, v_{n+2} + 1) \quad \text{for } n = 0, 1, \dots, 3028. \quad (*)$$

Proof. Let $0 \leq n \leq 3028$ and let $s = \min(v_n + 2, v_{n+2} + 1)$. Then $s \leq v_n + 2$ implies $2^s \mid 4a_n$ and $s \leq v_{n+2} + 1$ implies $2^s \mid 2a_{n+2}$. It follows that $a_{n+1} = 2a_{n+2} - 4a_n$ is also divisible by 2^s , hence $s \leq v_{n+1}$, which proves $(*)$.

Here are different ways of working out the induction argument that is crucial in the proofs.

Induction part, alternative 1.

Statement. For $k = 0, 1, \dots, 1010$, the terms $a_k, a_{k+1}, \dots, a_{3030-2k}$ are all divisible by 2^{2k} .

We reformulate the statement using the notation v_n as the largest integer such that 2^{v_n} divides a_n :

$$v_n \geq k \text{ for any } n \text{ satisfying } \left\lceil \frac{k}{2} \right\rceil \leq n \leq 3030 - k. \quad (**)$$

Here $\lceil x \rceil$ denotes the smallest integer not smaller than x .

Proof 1. We proceed by induction on k . For $k = 0$ the statement is obvious, so, for the inductive step, suppose that $a_k, a_{k+1}, \dots, a_{3030-2k}$ are all divisible by 2^{2k} . Apply Lemma 1 with

$$(a, b, c, d) = \left(\frac{a_{i-1}}{2^{2k}}, \frac{a_i}{2^{2k}}, \frac{a_{i+1}}{2^{2k}}, \frac{a_{i+2}}{2^{2k}} \right)$$

for $i = k+1, k+2, \dots, 3030-2k-2$. We obtain that $\frac{a_i}{2^{2k}}$ is divisible by 4 (and hence a_i is divisible by 2^{2k+2}) for $i = k+1, k+2, \dots, 3030-2k-2$.

This completes the induction. For $k = 1010$ we obtain that a_{1010} is divisible by 2^{2020} , and the solution is complete. $\square$

Remark. Notice that the this argument works in general, i.e. by replacing 1010 with n (and hence, 2020 with $2n$ and 3030 with $3n$). Then the claim is the following: if $a_0, a_1, \dots, a_{3n}$ are integers that satisfy the recursion in the problem, then a_n is divisible by 2^{2n} .

Proof 2. We will show by a two-step induction on $k \geq 0$ in $(**)$. Plugging in $k = 2020$ and $n = 1010$ will give the desired result.

The case $k = 0$ is trivial, since the v_n 's are non-negative. For the case $k = 1$, let $1 \leq n \leq 3029$, then $v_n \geq \min(v_{n-1} + 2, v_{n+1} + 1) \geq 1$. Suppose we have it proven for some $k - 1$ with $k \geq 1$. Let

$$\left\lceil \frac{k+1}{2} \right\rceil \leq n \leq 3030 - (k+1).$$

Then

$$\left\lceil \frac{k-1}{2} \right\rceil \leq n-1 \leq 3030 - (k-1) \text{ and } \left\lceil \frac{k}{2} \right\rceil \leq n+1 \leq 3030 - k.$$

By induction hypothesis we have $v_{n-1} \geq k-1$ and $v_{n+1} \geq k$. Then $v_n \geq \min(v_{n-1} + 2, v_{n+1} + 1) \geq k+1$, finishing the induction. $\square$

Proof 3. The notation is the same as above. We again use a two-step induction. We invoke Lemma 1 in the first step. Assume now that $2^{2k} \mid a_k, \dots, a_{3030-2k}$ for some $k \geq 1$. We claim that then $2^{2k+2} \mid a_{k+1}, \dots, a_{3030-2(k+1)}$. Notice first that any i on the interval $[k+1, 3030-2k-1]$ satisfies the equation $2a_{i+1} = a_i + 4a_{i-2}$, and so we have $2^{2k+1} \mid a_i$. Furthermore, if $i \in [k+1, 3030-2k-2]$, we have $2^{2k+2} \mid a_i$ since $2^{2k+1} \mid a_{i+1}$ and hence $2^{2k+2} \mid 2a_{i+1}$ and $2^{2k+2} \mid 4a_{i-1}$. $\square$

Induction part, alternative 2.

Statement. If $a_0, a_1, \dots, a_{3n}$ are integers that satisfy the recursion in the problem, then a_n is divisible by 2^{2n} . The problem statement follows when $n = 1010$.

Proof. The base case of the induction is exactly Lemma 1. Now, for the inductive step, suppose that the statement holds for $n = k-1$, and consider integers $a_0, a_1, \dots, a_{3k}$ that satisfy the recursion. By applying the induction hypothesis to the four sequences

$$(a_0, a_1, \dots, a_{3k-3}), (a_1, a_2, \dots, a_{3k-2}), (a_2, a_3, \dots, a_{3k-1}) \text{ and } (a_0, a_1, \dots, a_{3k}),$$

we find that a_{k-1}, a_k, a_{k+1} and a_{k+2} are all divisible by 2^{2k-2} . If we now apply Lemma 1 to $\frac{a_{k-1}}{2^{2k-2}}, \frac{a_k}{2^{2k-2}}, \frac{a_{k+1}}{2^{2k-2}}$, and $\frac{a_{k+2}}{2^{2k-2}}$, we find that $\frac{a_k}{2^{2k-2}}$ is divisible by 4, so a_k is divisible by 2^{2k} , as desired. $\square$

Induction part, alternative 3.

Statement. Given positive integers $a_0, a_1, a_2, \dots, a_{3k}$ such that $2a_{n+2} = a_{n+1} + 4a_n$ for $n = 0, 1, 2, \dots, 3k - 2$, then 2^{2k} divides at least one of the numbers $a_0, a_1, \dots, a_{3k}$.

Proof: The case $k = 1$ is obtained from Lemma. Suppose that for some $k \geq 1$, our claim is true for any sequence of $3k+1$ positive integers that satisfy similar defining relations. Then consider a sequence of $3(k+1) + 1 = 3k + 4$ positive integers $a_0, a_1, \dots, a_{3k+3}$ such that $2a_{n+2} = a_{n+1} + 4a_n$ for $n = 0, 1, 2, \dots, 3k + 1$. Then, we see that all numbers $a_1, a_2, \dots, a_{3k+2}$ are even, and so we may set $a_i = 2b_i$ for some positive integers b_i for $i = 1, 2, \dots, 3k + 2$. Then $2a_2 = a_1 + 4a_0$ becomes $2b_2 = b_1 + 2a_0$. Note that $2b_{n+2} = b_{n+1} + 4b_n$ for $n = 1, 2, \dots, 3k$. Hence the positive integers $b_1, b_2, \dots, b_{3k+1}$ are even, and so we may write $b_i = 2c_i$ for some positive integers c_i for $i = 1, 2, \dots, 3k + 1$. Then $2b_2 = b_1 + 2a_0$ becomes $2c_2 = c_1 + a_0$, which does not give us anything. However, we have $2c_{n+2} = c_{n+1} + 4c_n$ for $n = 1, 2, \dots, 3k - 1$. So the sequence of positive integers $c_1, c_2, \dots, c_{3k+1}$ is a sequence of $3k + 1$ positive integers that satisfy the defining relations. By the inductive hypothesis, 2^{2k} divides at least one of the numbers $c_1, c_2, \dots, c_{3k+1}$. Since $a_i = 4c_i$ for all $i = 1, 2, \dots, 3k + 1$, it follows that $4 \cdot 2^{2k} = 2^{2(k+1)}$ divides at least one of the numbers $a_1, \dots, a_{3k+1}$. This completes the induction, and hence the proof. $\square$

Induction part, alternative 4.

Statement. If $v_0, v_1, \dots, v_{3030}$ is a sequence of non-negative integers satisfying $(*)$, there must be a k such that $v_k \geq 2020$. In fact, we will show that $v_{1010} \geq 2020$.

The following proofs use Lemma 2.

Proof 1. We will show by induction on k that $v_n \geq 2k$ for $k \leq n \leq 3030 - 2k$ and $v_n \geq 2k + 1$ for $k + 1 \leq n \leq 3030 - 2k - 1$. For $k = 1010$, the first statement implies that $v_{1010} \geq 2020$. For $k = 0$ the first statement $v_n \geq 0$ is obvious, and the second statement follows using $(*)$: we have $v_n \geq v_{n+1} + 1 \geq 1$ for $1 \leq n \leq 3029$. Now suppose that the inductive hypothesis holds for $k = \ell$: we have $v_n \geq 2\ell$ for $\ell \leq n \leq 3030 - 2\ell$ and $v_n \geq 2\ell + 1$ for $\ell + 1 \leq n \leq 3030 - 2\ell - 1$. For the first statement, consider an n with $\ell + 1 \leq n \leq 3030 - 2\ell - 2$. Then using $(*)$ and the inductive hypothesis, we obtain

$$v_n \geq \min(v_{n-1} + 2, v_{n+1} + 1) \geq \min(2\ell + 2, 2\ell + 2) = 2\ell + 2$$

because $v_{n-1} \geq 2\ell$ (as $\ell \leq n - 1 \leq 3030 - 2\ell$) and $v_{n+1} \geq 2\ell + 1$ (as $\ell + 1 \leq n + 1 \leq 3030 - 2\ell - 1$). Similarly, for $\ell + 2 \leq n \leq 3030 - 2\ell - 3$ we find

$$v_n \geq \min(v_{n-1} + 2, v_{n+1} + 1) \geq \min(2\ell + 3, 2\ell + 3) = 2\ell + 3$$

because $v_{n-1} \geq 2\ell + 1$ (as $\ell + 1 \leq n - 1 \leq 3030 - 2\ell - 1$) and $v_{n+1} \geq 2\ell + 2$ (as $\ell + 1 \leq n \leq 3030 - 2\ell - 2$). This completes the induction. $\square$

Proof 2. The inequality $v_{1010} \geq \min(v_{1009} + 2, v_{1011} + 1)$ gives us two cases to consider: either $v_{1010} \geq v_{1009} + 2$ or $v_{1010} \geq v_{1011} + 1$.

Suppose first that $v_{1010} \geq v_{1009} + 2$. In this case we will show by induction on k that $v_{1010-k} \geq v_{1009-k} + 2$ for $0 \leq k \leq 1009$. The case $k = 0$ is assumed, so suppose that $v_{1010-\ell} \geq v_{1009-\ell} + 2$ holds for some $0 \leq \ell < 1009$. We obtain

$$v_{1009-\ell} \geq \min(v_{1010-\ell} + 1, v_{1008-\ell} + 2) \geq \min(v_{1009-\ell} + 3, v_{1008-\ell} + 2).$$

Because $v_{1009-\ell} < v_{1009-\ell} + 3$ we must have $v_{1009-\ell} \geq v_{1008-\ell} + 2$, completing the induction. We conclude that

$$v_{1010} \geq v_{1009} + 2 \geq v_{1008} + 4 \geq \dots \geq v_0 + 2020 \geq 2020.$$

In the second case, we show by induction on k that $v_{1010+k} \geq v_{1011+k} + 1$ for $0 \leq k \leq 2019$. Again, the base case $k = 0$ is assumed, so suppose that $v_{1010+\ell} \geq v_{1011+\ell} + 1$ holds for some $0 \leq \ell < 2019$. We obtain

$$v_{1011+\ell} \geq \min(v_{1010+\ell} + 2, v_{1012+\ell} + 1) \geq \min(v_{1011+\ell} + 3, v_{1012+\ell} + 1).$$

Because $v_{1011+\ell} < v_{1011+\ell} + 3$, we must have $v_{1011+\ell} \geq v_{1012+\ell} + 1$, completing the induction. We conclude that

$$v_{1010} \geq v_{1011} + 1 \geq v_{1012} + 2 \geq \dots \geq v_{3030} + 2020 \geq 2020,$$

as desired. $\square$

Alternative 5.

Statement. This solution is different from the other solutions. Shift the sequence so that it starts at a_{-1010} and ends at a_{2020} . We will show that a_0 is divisible by 2^{2020} . Consider a_0 . It either has at least as many factors 2 as $2a_1$, or at least as many factors 2 as $4a_{-1}$ (this follows from $2a_1 = a_0 + 4a_{-1}$). Consider the first case, so $e_2(a_0) \geq e_2(2a_1)$, where $e_2(n)$ denotes the highest exponent of 2 of an integer n . By multiplying the original recursion by 2^{n-1} , we note that $b_n = 2^n a_n$ satisfies the recursion $b_n = b_{n-1} + 8b_{n-2}$. Furthermore, we assumed that $e_2(b_0) \geq e_2(b_1)$. This implies that $e_2(b_n)$ is constant for $n \geq 1$. Furthermore, clearly $2^{2020} \mid b_{2020}$, so $2^{2020} \mid b_1$ and hence $2^{2020} \mid a_0$. The case where $e_2(a_0) \geq e_2(4a_{-1})$ is similar; we then look at $b_n = 4^n a_{-n}$ which satisfies $b_n = 8b_{n-2} - b_{n-1}$ and use that $2^{2020} \mid b_{1010}$. The rest of the argument is the same. $\square$

Problem 98. A permutation of the integers $1, 2, \dots, m$ is called *fresh* if there exists no positive integer $k < m$ such that the first k numbers in the permutation are $1, 2, \dots, k$ in some order. For example, if $m = 4$, then the permutation $(3, 1, 4, 2)$ is fresh, whereas the permutation $(2, 3, 1, 4)$ is not.

Let f_m be the number of fresh permutations of the integers $1, 2, \dots, m$.

Prove that $f_n \geq n \cdot f_{n-1}$ for all $n \geq 3$.

Solution 1. Let $\sigma = (\sigma_1, \dots, \sigma_{n-1})$ be a fresh permutation of the integers $1, 2, \dots, n-1$. We claim that for any $1 \leq i \leq n-1$, the permutation

$$\sigma^{(i)} = (\sigma_1, \dots, \sigma_{i-1}, n, \sigma_i, \dots, \sigma_{n-1})$$

is a fresh permutation of the integers $1, 2, \dots, n$. Indeed, let $1 \leq k \leq n-1$. If $k \geq i$ then we have $n \in \{\sigma_1^{(i)}, \dots, \sigma_k^{(i)}\}$, but $n \notin \{1, 2, \dots, k\}$. And if $k < i$ we have $k < n-1$, and $\{\sigma_1^{(i)}, \dots, \sigma_k^{(i)}\} = \{\sigma_1, \dots, \sigma_k\} \neq \{1, 2, \dots, k\}$, since σ is fresh. Moreover, it is easy to see that when we apply this construction to all fresh permutations of $1, 2, \dots, n-1$, we obtain $(n-1) \cdot f_{n-1}$ distinct fresh permutations of $1, 2, \dots, n$. Note that a fresh permutation of $1, 2, \dots, n-1$ cannot end in $n-1$, and hence none of the previously constructed fresh permutations of $1, 2, \dots, n$ will end in $n-1$ either. Therefore, we will complete the proof by finding f_{n-1} fresh permutations of $1, 2, \dots, n$ that end in $n-1$. To do this, let $\sigma = (\sigma_1, \dots, \sigma_{n-1})$ be a fresh permutation of $1, 2, \dots, n-1$ and let j be such that $\sigma_j = n-1$. Define

$$\sigma' = (\sigma_1, \dots, \sigma_{j-1}, n, \sigma_{j+1}, \dots, \sigma_{n-1}, n-1),$$

then clearly σ' is a permutation of $1, 2, \dots, n$ that ends in $n-1$. We show that σ' is fresh, so let $1 \leq k \leq n-1$. If $k \geq j$ then $n \in \{\sigma'_1, \dots, \sigma'_k\}$ but $n \notin \{1, 2, \dots, k\}$; if $k < j$, then $k < n-1$ and $\{\sigma'_1, \dots, \sigma'_k\} = \{\sigma_1, \dots, \sigma_k\} \neq \{1, 2, \dots, k\}$, since σ is fresh. So we have constructed f_{n-1} additional fresh permutations of $1, 2, \dots, n$ (which again are all different), and the total number f_n of fresh permutations of $1, 2, \dots, n$ must at least be $(n-1)f_{n-1} + f_{n-1} = nf_{n-1}$, as required. $\square$

Comment. A similar way to construct n fresh permutations of $1, 2, \dots, n$ for each fresh permutation σ of $1, 2, \dots, n-1$ is as follows: increase all entries of σ by 1, and then add the number 1 anywhere; all these permutations are fresh, except the ones where 1 is added at the front, which can be made fresh by swapping the 1 and the 2. It is again straightforward to check that we obtain $n \cdot f_{n-1}$ permutations that are fresh and distinct, although a little extra care is needed to account for the fact that we increased the entries of our original permutation.

Solution 2. Assuming $n \geq 3$, we construct $n \cdot f_{n-1}$ different fresh permutations. Consider a fresh permutation of the $n-1$ numbers $1, 3, 4, \dots, n$ (the number 2 has been removed), by which we mean a permutation $(x_1, \dots, x_{n-1})$ such that $x_1 \neq 1$ and $\{x_1, \dots, x_k\} \neq \{1, 3, 4, \dots, k+1\}$ for all k with $2 \leq k \leq n-2$. There are exactly f_{n-1} such permutations. By inserting 2 anywhere in such a permutation, i.e. before or after all entries, or between two entries x_i, x_{i+1} , we generate the following list of n distinct permutations of $1, 2, \dots, n$, which we claim are all fresh:

$$(2, x_1, \dots, x_{n-1}), \dots, (x_1, \dots, x_{i-1}, 2, x_i, \dots, x_{n-1}), \dots, (x_1, \dots, x_{n-1}, 2).$$

In order to verify freshness, suppose that some permutation above is not fresh, that is, for some $1 \leq k \leq n-1$, the first k elements are $1, \dots, k$. If $k = 1$ then the first element is 1; but the first element is either 2 or x_1 and $x_1 \neq 2$, so this is not possible. If $k \geq 2$ then the first k elements must contain 2 and $x_1, \dots, x_{k-1}$, so $\{x_1, \dots, x_{k-1}\} = \{1, 3, 4, \dots, k\}$, in contradiction to the fact that $(x_1, \dots, x_{n-1})$ is fresh. We have thus constructed $n \cdot f_{n-1}$ different fresh permutations, so $f_n \geq n \cdot f_{n-1}$. $\square$

Comment. Note that the construction from Solution 2 can also be performed by placing $n-1$ instead of 2, or indeed any k between 2 and $n-1$ (in which case it will work for $n \geq k+1$). Moreover, similar arguments can be made while choosing to work with non-fresh permutations rather than fresh permutations. Indeed, in order to show that $n! - f_n \leq n((n-1)! - f_{n-1})$ for all $n \geq 3$, which is equivalent to the problem statement, one can argue as follows. Given a non-fresh permutation $(\sigma_1, \dots, \sigma_n)$ of $1, 2, \dots, n$, erasing $n-1$ from it and changing n to $n-1$ yields a permutation which cannot be fresh. Moreover, each non-fresh permutation of $1, 2, \dots, n-1$ can be obtained in this way from at most n different non-fresh permutations of $1, 2, \dots, n$. This type of argument would essentially mirror the one carried out in Solution 2. $\square$

Solution 3. By considering all permutations of $1, 2, \dots, n$ and considering the smallest k for which the first k numbers are $1, 2, \dots, k$ in some order, one can deduce the recursion

$$\sum_{k=1}^n (n-k)! \cdot f_k = n!,$$

which holds for any $n \geq 1$. Hence, if $n \geq 3$ we have

$$\begin{aligned} 0 &= n! - (n+1)(n-1)! + (n-1)(n-2)! \\ &= \sum_{k=1}^n (n-k)! \cdot f_k - (n+1) \sum_{k=1}^{n-1} (n-1-k)! \cdot f_k + (n-1) \sum_{k=1}^{n-2} (n-2-k)! \cdot f_k \\ &= f_n - nf_{n-1} + \sum_{k=1}^{n-2} (n-2-k)! \cdot ((n-k)(n-k-1) - (n+1)(n-k-1) + (n-1)) \cdot f_k \\ &= f_n - nf_{n-1} + \sum_{k=1}^{n-2} (n-2-k)! \cdot k(k+2-n) \cdot f_k \\ &\leq f_n - nf_{n-1}, \end{aligned}$$

which rewrites as $f_n \geq n \cdot f_{n-1}$.

Solution 4. We first show that for any $n \geq 2$ we have

$$f_n = \sum_{k=1}^{n-1} k(n-1-k)! \cdot f_k.$$

To deduce this relation, we imagine obtaining permutations of $1, 2, \dots, n$ by inserting n into a permutation of the numbers $1, 2, \dots, n-1$. Specifically, let $(\sigma_1, \dots, \sigma_{n-1})$ be any permutation of $1, 2, \dots, n-1$, and let $1 \leq k \leq n-1$ be minimal with $\{\sigma_1, \dots, \sigma_k\} = \{1, 2, \dots, k\}$. Note that there are $(n-1-k)! \cdot f_k$ such permutations. Furthermore, inserting n in this permutation will give a fresh permutation if and only if n is inserted before σ_k , so there are k options to do so. Consequently, if $n \geq 3$,

$$\begin{aligned} f_n &= \sum_{k=1}^{n-1} k(n-1-k)! \cdot f_k \\ &= (n-1)f_{n-1} + \sum_{k=1}^{n-2} k(n-1-k)! \cdot f_k \\ &\geq (n-1)f_{n-1} + \sum_{k=1}^{n-2} k(n-2-k)! \cdot f_k \\ &= (n-1)f_{n-1} + f_{n-1} \\ &= n \cdot f_{n-1}, \end{aligned}$$

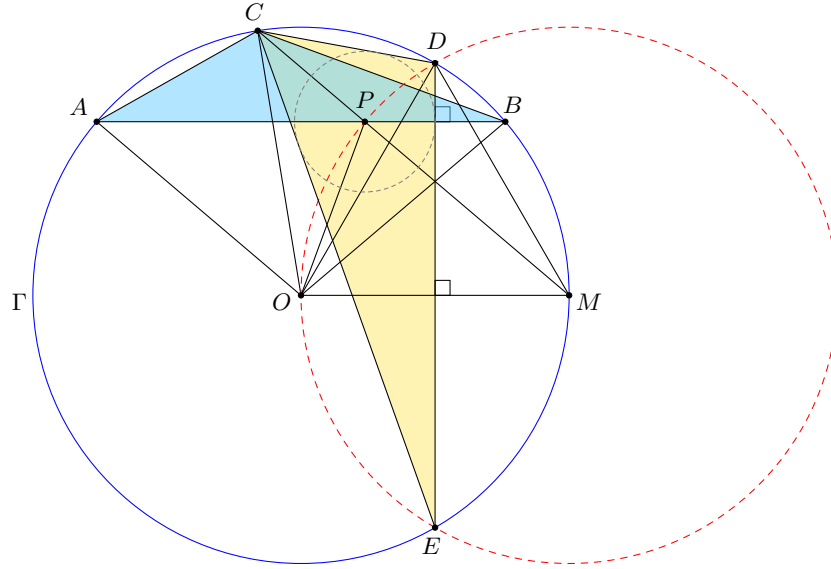
completing the proof. $\square$

Comment. Fresh permutations are known as *indecomposable permutations* or *irreducible permutations* in the literature. The problem asks to prove that the probability that a randomly chosen permutation of $1, 2, \dots, n$ is indecomposable is a non-decreasing function of n . In fact, it turns out that this probability goes to 1 as $n \rightarrow \infty$; for large n , almost all permutations of $1, 2, \dots, n$ are indecomposable. More can be found in: Y. Koh & S. Ree. Connected permutation graphs. *Discrete Mathematics* **307** (21):2628-2635, 2007.

Problem 99. Consider the triangle ABC with $\angle BCA > 90^\circ$. The circumcircle Γ of ABC has radius R . There is a point P in the interior of the line segment AB such that $PB = PC$ and the length of PA is R . The perpendicular bisector of PB intersects Γ at the points D and E .

Prove that P is the incentre of triangle CDE .

Solution 1.



The angle bisector of $\angle ECD$ intersects the circumcircle of $\triangle CDE$ (which is Γ) at the midpoint M of $\widehat{DBE}$. It is well-known that the incentre is the intersection of the angle bisector segment CM and the circle with centre at M and passing through D, E . We will verify this property for P .

By the conditions we have $AP = OA = OB = OC = OD = OE = R$. Both lines OM and APB are perpendicular to ED , therefore $AP \parallel OM$; in the quadrilateral $AOMP$ we have $AP = OA = OM = R$ and $AP \parallel OM$, so $AOMP$ is a rhombus and its fourth side is $PM = R$. In the convex quadrilateral $OMBP$ we have $OM \parallel PB$, so $OMBP$ is an isosceles trapezoid; the perpendicular bisector of its bases MO and PB coincide. From this symmetry we obtain $MD = OD = R$ and $ME = OE = R$. (Note that $\triangle OEM$ and $\triangle OMD$ are equilateral.) We already have $MP = MD = ME = R$, so P indeed lies on the circle with center M and passing through D, E . (Notice that this circle is the reflection of Γ about DE .)

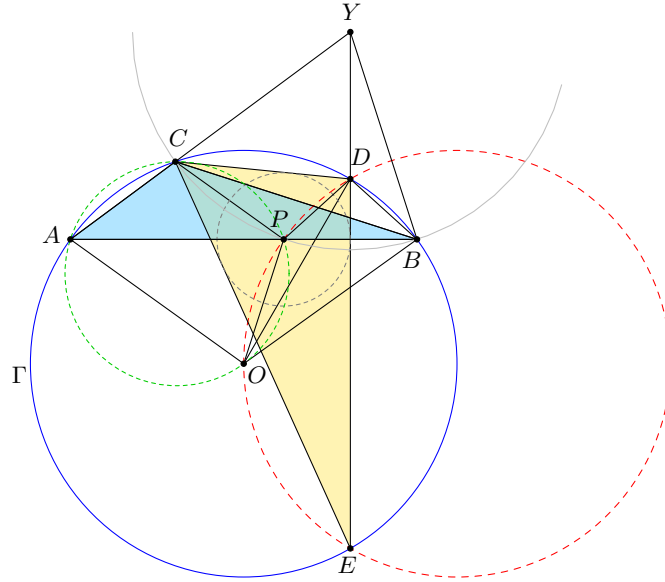
From $PB = PC$ and $OB = OC$ we know that B and C are symmetric about OP ; from the rhombus $AOMP$ we find that A and M are also symmetric about OP . By reflecting the collinear points B, P, A (with P lying in the middle) we get that C, P, M are collinear (and P is in the middle). Therefore P lies on the line segment CM . $\square$

Solution 2.

Let M be the second intersection of CP with Γ . Using the power of the point P in the circle Γ and the fact that $PB = PC$, we find that $PM = PA = R$. The quadrilateral $AOMP$ has four sides of equal length, so it is a rhombus and in particular $OM \parallel AP$. This proves that $OMBP$ is a trapezoid, and because the diagonals PM and OB have equal length. In fact it is an isosceles trapezoid. Because of that, DE is not only the perpendicular bisector of PB , but also of OM .

In particular we have $MD = MP = MO = ME = R$, which proves that M is the midpoint of $\widehat{DBE}$ and P belongs to the circle with center M going through D and E . These properties, together with the fact that C, P, M are collinear determine uniquely the incentre of $\triangle CDE$. $\square$

Solution 3.



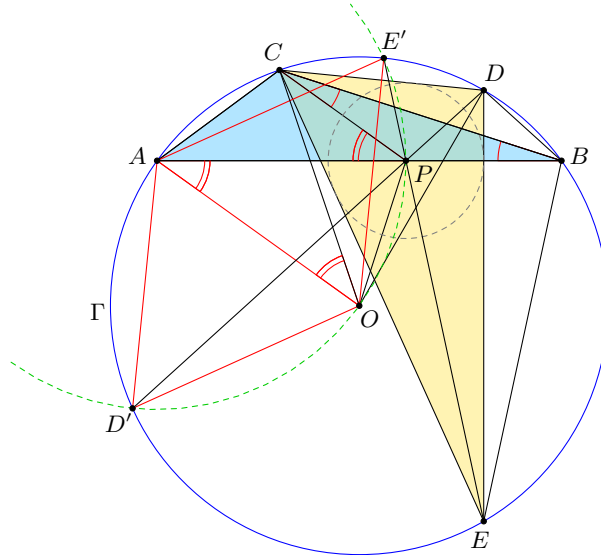
Let Y be the circumcentre of $\triangle BPC$. Then from $YB = YP$ it follows that Y lies on DE (we assume D lies in between Y and E), and from $YB = YC$ it follows that Y lies on OP , where O is the centre of Γ .

From $\angle AOC = 2\angle ABC = \angle APC$ (because $\angle PBC = \angle PCB$) we deduce that A, O, P and C are concyclic, and from $AP = R$ it follows that $AOPC$ is an isosceles trapezoid. We now find that $\angle YCP = \angle YPC = 180^\circ - \angle OPC = 180^\circ - \angle ACP$, so Y lies on AC .

Power of a point gives $YO \cdot YP = YC \cdot YA = YD \cdot YE$, so D, P, O and E are concyclic. It follows that $2\angle DAE = \angle DOE = \angle DPE = \angle DBE = 180^\circ - \angle DAE$, so $\angle DAE = 60^\circ$. We can now finish the proof by angle chasing.

From $AB \perp DE$ we have $\angle AOD + \angle BOE = 180^\circ$ and from $\angle DOE = 2\angle DAE = 120^\circ$ it follows that $\angle BOD + \angle BOE = 120^\circ$. It follows that $\angle AOD - \angle BOD = 180^\circ - 120^\circ = 60^\circ$. Let $\angle OAB = \angle OBA = 2\beta$; then $\angle AOD + \angle BOD = \angle AOB = 180^\circ - 4\beta$. Together with $\angle AOD - \angle BOD = 60^\circ$, this yields $\angle AOD = 120^\circ - 2\beta$ and $\angle BOD = 60^\circ - 2\beta$. We now find $\angle AED = \frac{1}{2}\angle AOD = 60^\circ - \beta$, which together with $\angle DAE = 60^\circ$ yields $\angle ADE = 60^\circ + \beta$. From the isosceles trapezoid $AOPC$ we have $\angle CDA = \angle CBA = \frac{1}{2}\angle CPA = \frac{1}{2}\angle PAO = \beta$, so $\angle CDE = \angle CDA + \angle ADE = \beta + 60^\circ + \beta = 60^\circ + 2\beta$. From $\angle BOD = 60^\circ - 2\beta$ we deduce that $\angle BED = 30^\circ - \beta$; together with $\angle DBE = 120^\circ$ this yields $\angle EDB = 30^\circ + \beta$. We now see that $\angle PDE = \angle BDE = 30^\circ + \beta = \frac{1}{2}\angle CDE$, so P is on the angle bisector of $\angle CDE$. Similarly, P lies on the angle bisector of $\angle CED$. Therefore P is the incentre of $\triangle CDE$. $\square$

Solution 4.



Let DP, EP meet Γ again at D', E' , respectively. Observe that $\triangle APD' \sim \triangle DPB$, and $\triangle APE' \sim \triangle EPB$. Moreover, because $DP = DB$ and $EP = EB$, they are all isosceles and it follows that E', O, P, D' lie on a circle with centre A . In particular, $\triangle AOD'$ and $\triangle AOE'$ are equilateral triangles. Angle chasing gives

$$\angle CDP = \angle CDD' = \frac{1}{2}\angle COD' = \frac{1}{2}(60^\circ + \angle COA)$$

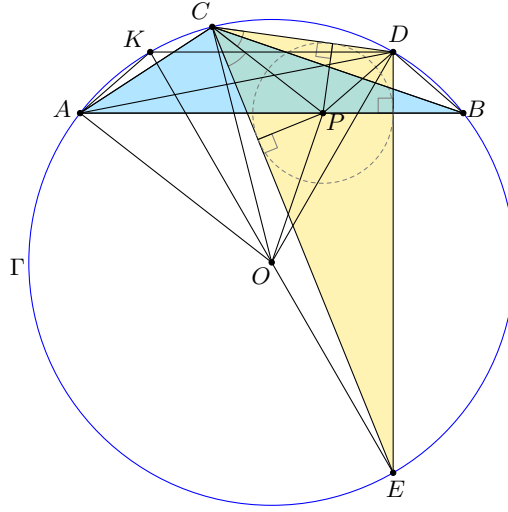
and

$$\angle EDP = \angle EDD' = \angle EE'D' = \angle PE'D' = \frac{1}{2}\angle PAD' = \frac{1}{2}(60^\circ + \angle PAO).$$

Similarly we prove $\angle CEP = \frac{1}{2}(60^\circ - \angle COA)$ and $\angle DEP = \frac{1}{2}(60^\circ - \angle PAO)$ so if we can prove that $\angle COA = \angle PAO$, we will have proven that P belongs to the angle bisector of $\angle CED$ and to the angle bisector of $\angle CDE$, which is enough to prove that P is the incentre of $\triangle CDE$.

Let $\beta = \angle ABC = \angle PCB$. We have $\angle APC = 2\beta$ and $\angle AOC = 2\beta$, so $AOPC$ is an inscribed quadrilateral. Moreover, since the diagonals AP and CO have equal length, this is actually an isosceles trapezoid, and hence $\angle PAO = \angle CPA = \angle COA$ which concludes the proof. $\square$

Solution 5.



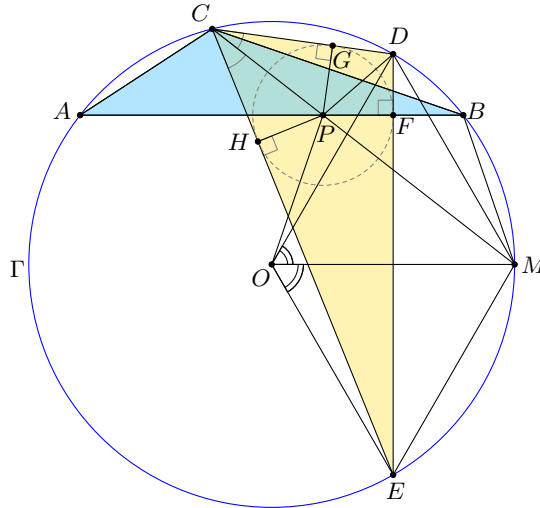
Without loss of generality we assume that D and C are in the same half-plane regarding line AB . Since $PC = BP$ and $ABCD$ is inscribed quadrilateral we have $\angle PCB = \angle CBP = \angle CEA = \alpha$. As in the other solutions, $AOPC$ is an isosceles trapezoid and $2\alpha = \angle CPA = \angle PCO$. Let K be intersection of EO and Γ . Then $\angle KDE = 90^\circ$, $AB \parallel DK$ and $KDBA$ is isosceles trapezoid. We obtain $DP = BD = AK$, which implies that $DPAK$ is a parallelogram and hence $DK = PA = R = OD = OK$. We see that $\triangle DOK$ is an equilateral triangle. Then $\angle ECD = \angle EKD = 60^\circ$.

Further we prove that PC bisects $\angle ECD$ using $\angle DCB = \angle DKB = \angle KDA$ (from isosceles trapezoid $DKAB$) and that $\angle KEC = \angle OEC = \angle OCE$ (because $\triangle OCE$ is isosceles):

$$\angle DCP = \angle DCB + \angle BCP = \angle KDA + \alpha = \angle CEA - \angle KEC + \alpha = 2\alpha - \angle OCE = \angle PCE$$

Further by $\angle DCP = \angle PCE = \frac{1}{2}\angle ECD = 30^\circ$, distances between P and sides of $\triangle CDE$ are $\frac{1}{2}PC = \frac{1}{2}PB$ (as ratio between the legs and the hypotenuse in right triangle with angles 60° and 30°). Therefore P is the incentre of $\triangle CDE$. $\square$

Solution 6.



Let F, G, H be the projections of P on the sides DE, DC, CE , respectively. If P is indeed the incentre, then the three line segments PF, PG, PH have the same length. This means that the problem is equivalent to proving that $PG = PH = PF = \frac{1}{2}PB = \frac{1}{2}PC$ and thus trigonometry in the right-angled triangles CPG and CPH tells us that it is enough to prove that $\angle DCP = \angle ECP = 30^\circ$.

We introduce the point M as the second intersection of the line CP with Γ . Because O is the centre of Γ we can reduce the problem to proving that $\angle MOD = \angle MOE = 60^\circ$, or equivalently that $\triangle MOD$ and $\triangle MOE$ are equilateral triangles. This last condition is equivalent to M being the reflection of O on the line DE . Following the chain of equivalences, we see therefore that in order to solve the problem it is enough to prove that M is the reflection of O on DE . We prove this property as in Solution 2, using the fact that $OMBP$ is an isosceles trapezoid. $\square$

Solution 7. Assume AB is parallel to the horizontal axis, and that Γ is the unit circle. Write $f(\theta)$ for the point $(\cos \theta, \sin \theta)$ on Γ . As in Solution 3, assume that $\angle AOB = 180^\circ - 4\beta$; then we can take $B = f(2\beta)$ and $A = f(180^\circ - 2\beta)$. As in Solution 3, we observe that $AOPC$ is an isosceles trapezoid, which we use to deduce that $\angle ABC = \frac{1}{2}\angle APC = \frac{1}{2}\angle OAB = \beta$. We now know that $C = f(180^\circ - 4\beta)$.

The point P lies on AB with $AP = R = 1$, so $P = (\cos(180^\circ - 2\beta) + 1, \sin(2\beta)) = (1 - \cos(2\beta), \sin(2\beta))$. The midpoint of BP therefore has coordinates $(\frac{1}{2}, \sin(2\beta))$, so D and E have x -coordinate $\frac{1}{2}$. Without loss of generality, we take $D = f(60^\circ)$ and $E = f(-60^\circ)$.

We have now obtained coordinates for all points in the problem, with one free parameter β . To show that P is the incentre of $\triangle CDE$, we will show that P lies on the bisector of $\angle CDE$; analogously, one can show that P lies on the bisector of $\angle CED$. The bisector of $\angle CDE$ passes through the midpoint M of the $\widehat{CE}$ not containing DE ; because $C = f(180^\circ - 4\beta)$ and $E = f(-60^\circ)$, we have $M = f(240^\circ - 2\beta)$.

It remains to show that $P = (1 - \cos(2\beta), \sin(2\beta))$ lies on the line connecting the points $D = (\cos 60^\circ, \sin 60^\circ)$ and $M = (\cos(240^\circ - 2\beta), \sin(240^\circ - 2\beta)) = (-\cos(60^\circ - 2\beta), -\sin(60^\circ - 2\beta))$. The equation for the line DM is

$$(y + \sin(60^\circ - 2\beta))(\cos 60^\circ + \cos(60^\circ - 2\beta)) = (\sin 60^\circ + \sin(60^\circ - 2\beta))(x + \cos(60^\circ - 2\beta)),$$

which, using the fact that $\cos 60^\circ + \cos(60^\circ - 2\beta) = 2 \cos \beta \cos(60^\circ - \beta)$ and $\sin 60^\circ + \sin(60^\circ - 2\beta) = 2 \cos(\beta) \sin(60^\circ - \beta)$, simplifies to

$$(y + \sin(60^\circ - 2\beta)) \cos(60^\circ - \beta) = (x + \cos(60^\circ - 2\beta)) \sin(60^\circ - \beta).$$

Because $\cos(60^\circ - 2\beta) \sin(60^\circ - \beta) - \sin(60^\circ - 2\beta) \cos(60^\circ - \beta) = \sin \beta$, this equation further simplifies to

$$y \cos(60^\circ - \beta) - x \sin(60^\circ - \beta) = \sin \beta.$$

Plugging in the coordinates of P , i.e., $x = 1 - \cos(2\beta)$ and $y = \sin(2\beta)$, shows that P is on this line: for this choice of x and y , the left-hand side equals $\sin(60^\circ + \beta) - \sin(60^\circ - \beta) = 2 \cos 60^\circ \sin \beta$, which is indeed equal to $\sin \beta$. So P lies on the bisector DM of $\angle CDE$, as desired. $\square$

Solution 8. Let Γ be the unit circle on the complex plane and the small letters as usual are complex numbers representing the given points. Let AB be parallel with the real line and $0 < \varphi = \arg b < 90^\circ$. Then

$$|b| = 1, \quad a = -\bar{b}, \quad p = a + 1 = 1 - \bar{b}.$$

From $\operatorname{Re} d = \operatorname{Re} e = \operatorname{Re} \frac{p+b}{2} = \frac{1}{2}$ we get that $d = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $e = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ are conjugate 6th roots of unity; $d^3 = e^3 = -1$, $d + e = 1$, $d^2 = -e$, $e^2 = -d$ etc.

Point C is the reflection of B in line OP . From $\arg p = \arg(1 - \bar{b}) = \frac{1}{2}(180^\circ - \varphi)$, we can get $\arg c = 2 \arg p - \arg b = 180^\circ - 2\varphi$, so $c = -\bar{b}^2$. Now we can verify that EP bisects $\angle CED$. This happens if and only if $(p - e)^2(\bar{c} - \bar{e})(\bar{d} - \bar{e})$ is real. Since $\bar{d} - \bar{e} = -i\sqrt{3}$, this is equivalent to showing that $\operatorname{Re}(p - e)^2(\bar{c} - \bar{e}) = 0$. Now

$$\begin{aligned} (p - e)^2(\bar{c} - \bar{e}) &= (1 - \bar{b} - e)^2(-b^2 - d) \\ &= (d - \bar{b})^2(-b^2 - d) \\ &= -|b|^4 + 2d|b|^2b - d^2b^2 - d\bar{b}^2 + 2d^2\bar{b} - d^3 \\ &= -1 - 2db + \bar{d}b^2 - d\bar{b}^2 - 2d\bar{b} + 1 \\ &= -2(db - \bar{d}\bar{b}) + (\bar{d}b^2 - d\bar{b}^2), \end{aligned}$$

has real part equal to zero. Similarly DP bisects $\angle EDC$. $\square$

Problem 100. Let $m > 1$ be an integer. A sequence $a_1, a_2, a_3, \dots$ is defined by $a_1 = a_2 = 1$, $a_3 = 4$, and for all $n \geq 4$,

$$a_n = m(a_{n-1} + a_{n-2}) - a_{n-3}.$$

Determine all integers m such that every term of the sequence is a square.

Solution 1. The answers are $m = 2$ and $m = 10$.

Consider an integer $m > 1$ for which the sequence defined in the problem statement contains only perfect squares. We shall first show that $m - 1$ is a power of 3.

Suppose that $m - 1$ is even. Then $a_4 = 5m - 1$ should be divisible by 4 and hence $m \equiv 1 \pmod{4}$. But then $a_5 = 5m^2 + 3m - 1 \equiv 3 \pmod{4}$ cannot be a square, a contradiction. Therefore $m - 1$ is odd.

Suppose that an odd prime $p \neq 3$ divides $m - 1$. Note that $a_n - a_{n-1} \equiv a_{n-2} - a_{n-3} \pmod{p}$. It follows that modulo p the sequence takes the form $1, 1, 4, 4, 7, 7, 10, 10, \dots$; indeed, a simple induction shows that $a_{2k} \equiv a_{2k-1} \equiv 3k - 2 \pmod{p}$ for $k \geq 1$. Since $\gcd(p, 3) = 1$ we get that the sequence $a_n \pmod{p}$ contains all the residues modulo p , a contradiction since only $(p + 1)/2$ residues modulo p are squares. This shows that $m - 1$ is a power of 3.

Let h, k be integers such that $m = 3^k + 1$ and $a_4 = h^2$. We then have $5 \cdot 3^k = (h - 2)(h + 2)$. Since $\gcd(h - 2, h + 2) = 1$, it follows that $h - 2$ equals either $1, 3^k$ or 5 , and $h + 2$ equals either $5 \cdot 3^k, 5$ or 3^k , respectively. In the first two cases we get $k = 0$ and in the last case we get $k = 2$. This implies that either $m = 2$ or $m = 10$.

We now show the converse. Suppose that $m = 2$ or $m = 10$. Let $t = 1$ or $t = 3$ so that $m = t^2 + 1$. Let $b_1, b_2, b_3, \dots$ be a sequence of integers defined by $b_1 = 1, b_2 = 1, b_3 = 2$, and

$$b_n = tb_{n-1} + b_{n-2},$$

for all $n \geq 4$. Clearly, $a_n = b_n^2$ for $n = 1, 2, 3$. Note that if $m = 2$ then $a_4 = 9$ and $b_4 = 3$, and if $m = 10$ then $a_4 = 49$ and $b_4 = 7$. In both the cases we have $a_4 = b_4^2$.

If $n \geq 5$ then we have

$$b_n^2 + b_{n-3}^2 = (tb_{n-1} + b_{n-2})^2 + (b_{n-1} - tb_{n-2})^2 = (t^2 + 1)(b_{n-1}^2 + b_{n-2}^2) = m(b_{n-1}^2 + b_{n-2}^2).$$

Therefore, it follows by induction that $a_n = b_n^2$ for all $n \geq 1$. This completes the solution. □

Solution 2. The answers are $m = 2$ and $m = 10$.

We present an alternative proof that $m = 2$ and $m = 10$ are the only possible values of m with the required property.

Note that

$$\begin{aligned}a_4 &= 5m - 1, \\a_5 &= 5m^2 + 3m - 1, \\a_6 &= 5m^3 + 8m^2 - 2m - 4.\end{aligned}$$

Since a_4 and a_6 are squares, so is a_4a_6 . We have

$$4a_4a_6 = 100m^4 + 140m^3 - 72m^2 - 72m + 16.$$

Notice that

$$\begin{aligned}(10m^2 + 7m - 7)^2 &= 100m^4 + 140m^3 - 91m^2 - 98m + 49 < 4a_4a_6, \\(10m^2 + 7m - 5)^2 &= 100m^4 + 140m^3 - 51m^2 - 70m + 25 > 4a_4a_6,\end{aligned}$$

so we must have

$$4a_4a_6 = (10m^2 + 7m - 6)^2 = 100m^4 + 140m^3 - 71m^2 - 84m + 36.$$

This implies that $m^2 - 12m + 20 = 0$, so $m = 2$ or $m = 10$. □